

WASP-11b

R-1025

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_161104 · created 2026-07-30T05:13:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457696.7212 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1246 +/- 0.0092
Transit depth (Rp/Rs)^2	1.55 +/- 0.23 [%]
Orbital Inclination (inc)	87.53 +/- 0.74
Airmass coefficient 1 (a1)	1.164 +/- 0.011
Airmass coefficient 2 (a2)	-0.1443 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	None
Optimal Aperture	4.3
Optimal Annulus	11.28
Transit Duration (day)	0.0933 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1246 ± 0.0092 vs 0.131 (deviation 0.7σ)
Duration fit vs published	0.0933 d vs 0.1075 d (deviation 1.65σ)
Mid-time O–C	-3.3 min
Depth SNR	13.5
Residual scatter	0.9 %

Light curve

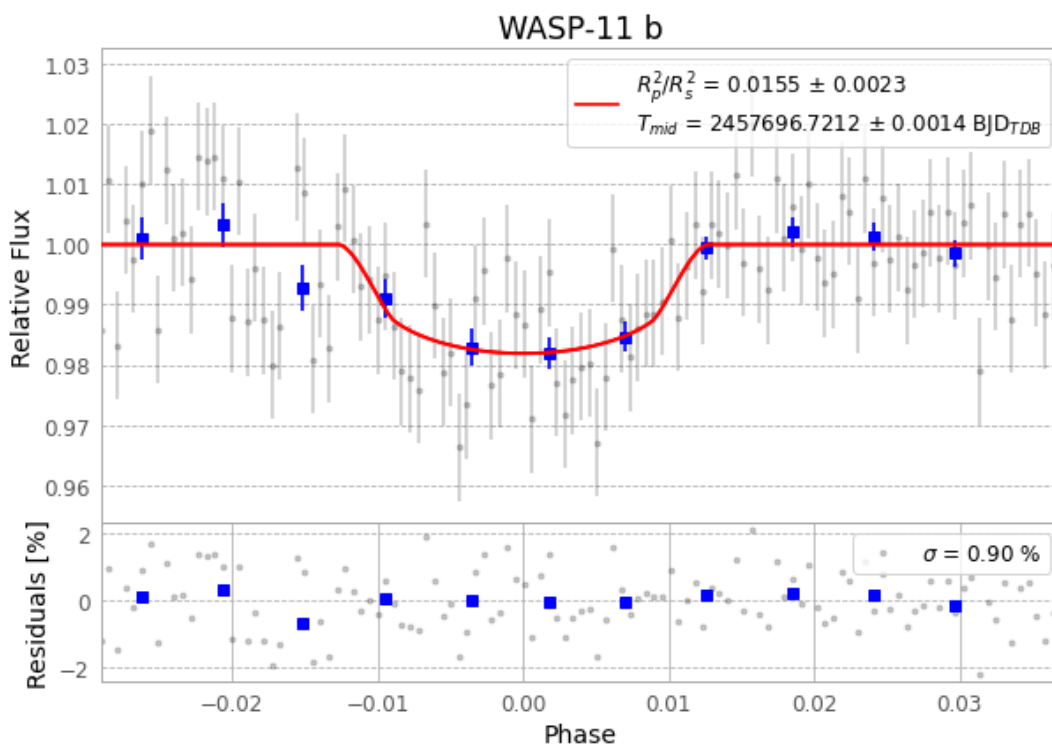


Figure 1 — FinalLightCurve_WASP-11 b_2016-11-03.png (SHA-256 9e1d0ab73c665446...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7592ca1fd75a994a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f8439d0

Data provenance

119 FITS frames (78 MB), HATP-10161104023312.FITS → HATP-10161104082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-161104.log (9e5066d01a9c...), FinalLightCurve_WASP-11 b_2016-11-03.png (9e1d0ab73c66...), FinalLightCurve_WASP-11 b_2016-11-03.csv (93b3e933979f...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 05:13Z.